

Certification of Dynamical System Models using End-to-End Distributionally-Robust Uncertainty Quantification

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We present `OptimalUncertaintyQuantification.jl`: A Julia tool for end-to-end distributionally robust uncertainty quantification of static and dynamical systems models. It provides an ergonomic symbolic interface for modeling the system as well as principled inclusion of data and knowledge about uncertainties. Furthermore, it automatically transforms the certification problem into a computationally tractable optimization problem solvable by integrated solvers. We demonstrate this tool on the benchmark Moore-Greitzer Ordinary Differential Equation model and provide an iterative workflow for model-based systems designers to meet safety certification targets.

I. Nomenclature

x	=	Real-valued random variable
x_{LB}	=	Lower bound of random variable x
x_{UB}	=	Upper bound of random variable x
μ_x	=	Measure denoting the law of the random variable x
$(\mu_x \times \mu_y)$	=	Product measure of μ_x and μ_y
$\mathcal{A}$	=	Admissible set of (product) measures that are consistent with available knowledge or data
$\mathbb{E}[\varphi(x) \mid \mu_x]$	=	Expectation of a function of x i.e., the Lebesgue integral of $\varphi(x)$ wrt. μ_x
$\mathbb{P}_{\mu_x}[g(x) \in \mathcal{X}]$	=	Probability that a function of x takes on a value in the event set $\mathcal{X}$
$\sup$	=	Supremum

II. Introduction

The field of Uncertainty Quantification (UQ) is concerned with explicitly accounting for the influence of various sources of uncertainty (e.g., input parameters, model structure, numerical simulation errors) on the predictions made by a model. The most-widely used technique of performing a Monte Carlo simulation relies on making a priori assumptions on the probability distributions of the uncertain parameters. In addition, Monte Carlo simulation often requires drawing a large number of samples and propagating them through the model which may be computationally expensive. Furthermore, making these unwarranted assumptions can lead to brittle and incorrect results [1]. This is particularly important when making certification/decertification decisions for safety-critical aerospace applications.

In order to address this issue, Owhadi et al. [2] pioneered the Optimal Uncertainty Quantification (OUQ) framework as an approach that avoids making probability distribution assumptions but instead introduces partial information and data about the model in a systematic way. OUQ then casts the UQ problem as an optimization problem searching over the space of probability distributions with constraints encoding the known information (such as moments or bounds on

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the random variables). In this way, uncertainty quantification is performed in a distributionally-robust way since the worst-case probability distribution is considered during the certification process.

In this work, we present `OptimalUncertaintyQuantification.jl`: A Julia package for end-to-end distributionally robust uncertainty quantification of static and dynamic systems models. We formally present the OUQ problem and illustrate the proposed approach on the Moore-Greitzer model [3] - a standard benchmark used in reachability analysis. We present the user interface for ergonomic formulation and solution of the OUQ problem leveraging the rich Julia technical computing ecosystem. We illustrate how the OUQ approach can be used for certification of dynamical system models. Finally, we present features under development and future work.

III. Problem Description

In the OUQ framework, knowledge and (experimental) data about the statistics of the uncertain random variables are introduced in a direct principled way. Typically, this knowledge takes the form of bounds on the random variables, moments information and other expectation information. This information is compiled into an “Admissible Set” - $\mathcal{A}$ which denotes the set of probability distributions that is consistent with all available knowledge and data.

In Eq.(A.S), $\mathcal{A}$ is the admissible set shown for a model with a single random variable x which takes on values within bounds $[x_{LB}, x_{UB}]$ where data is available for N expectation constraints. Thus, $\mathcal{A}$ defines the set of all possible measures (probability distributions) μ_x that are consistent with data on these generalized moment constraints.

$$\mathcal{A} = \left\{ \mu_x \left| \begin{array}{l} x \in [x_{LB}, x_{UB}] \subseteq \mathbb{R}, \\ \mathbb{E} [\varphi_j(x) \mid \mu_x] \leq c_j, \quad j = 1, \dots, N \end{array} \right. \right\} \quad (\text{A.S})$$

Problem OUQ presents the Optimal Uncertainty Quantification problem: The objective is determine the minimum and maximum probability that the relevant observable function g of the random variable takes on a value in the event set $\mathcal{X}$. We note that the true underlying measure of the model μ_{true} is never computed - instead rigorous lower and upper bounds (i.e., minimum and maximum probabilities of the event occurring) are computed. These computed bounds are then used for certification: For example, if $\mathcal{X}$ denotes an unsafe event set, then the model can be certified as safe if the computed maximum probability of the event occurring (right-hand side) is below a specified safety threshold.

$$\underbrace{\inf_{\mu_x \in \mathcal{A}} \mathbb{P}_{\mu_x}[g(x) \in \mathcal{X}]}_{\text{minimum probability}} \leq \mathbb{P}_{\mu_{true}}[g(x) \in \mathcal{X}] \leq \underbrace{\sup_{\mu_x \in \mathcal{A}} \mathbb{P}_{\mu_x}[g(x) \in \mathcal{X}]}_{\text{maximum probability}}, \quad (\text{OUQ})$$

IV. Motivating Example

We illustrate the OUQ framework on the Moore-Greitzer model [3]. This is an Ordinary Differential Equation (ODE) with 2 state variables x and y as shown in the equation below:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -y - \frac{3}{2}x^2 - \frac{1}{2}x^3 - \frac{1}{2} \\ 3x - y \end{bmatrix} \quad (\text{ODE})$$

The initial conditions x_0, y_0 are random variables. We assume these two random variables are mutually independent, thus the joint law of the pair is the product measure $\mu_{x_0} \times \mu_{y_0}$. Further, let admissible set $\mathcal{A}_1$ denote available knowledge and data about the problem.

$$\mathcal{A}_1 = \left\{ \mu_{x_0} \times \mu_{y_0} \left| \begin{array}{l} (x_0, y_0) \in [0.8, 1.2]^2 \subseteq \mathbb{R}^2, \\ \mathbb{E}[x_0 \mid \mu_{x_0}] = 1.0, \\ \mathbb{E}[y_0 \mid \mu_{y_0}] = 1.0 \end{array} \right. \right\} \quad (\text{A1})$$

At one particular point in time $t = 2.5$ seconds, let there exist an unsafe region $y \leq -3.1$. The goal is to determine the worst-case probability that the observable state y enters this unsafe region. The model is certified if this worst-case probability is below some allowable threshold of say 2% and it is decertified otherwise. To solve this certification problem, we formulate OUQ problem OUQ-MG which searches over the (infinite-dimensional) space of admissible

product measures: The objective function value corresponding to the extremal measure is the required worst-case probability of entering the unsafe region.

$$\begin{aligned} & \sup_{(\mu_{x_0} \times \mu_{y_0}) \in \mathcal{A}_1} \mathbb{P}_{(\mu_{x_0} \times \mu_{y_0})}[y[2.5] \leq -3.1] \\ & \text{s.t.} \quad \text{Eqns. ODE and A1,} \end{aligned} \tag{OUQ-MG}$$

Fig.(1) presents the reachable set of the phase space and shows this unsafe region for the observable state y at time $t = 2.5$ s.

V. Methodology

The `OptimalUncertaintyQuantification.jl` package provides a comprehensive tool for formulation and tractable solution of the OUQ problem. The following capabilities are highlighted:

- An ergonomic problem formulation interface. Subpackage `SymbolicOUQ.jl` provides useful macros (such as `@random_variables`) together with the expectation ($\mathbb{E}$) and probability ($\mathbb{P}$) operators. Support is provided for explicit specification of dependence/independence between random variables.
- Integration with the Julia differentiable modeling and simulation (SciML) ecosystem [4]. Native support for composable symbolic modeling of static and dynamical systems using `Symbolics.jl` [5] and `ModelingToolkit.jl` [6].
- Transformation algorithms that leverage reduction theorems to reformulate Problem OUQ-MG as a tractable finite-dimensional optimization problem. Currently, two transformation algorithms are implemented in `OptimalUncertaintyQuantification.jl`: `WinklerExtremalMeasures` and `StengerCanonicalMoments` as discussed in Section V.A.
- Integration with the Julia Reachability Analysis and Validated Numerics ecosystem. This is discussed in Section V.B.
- Integration with the Julia Optimization ecosystems. `OptimalUncertaintyQuantification.jl` provides an interface to both the `Optimization.jl` and `JuMP.jl` ecosystems as well as their corresponding solvers.
- A native configurable branch-and-bound global optimization solver as discussed in Section V.C.

A. Reduction Theorems

The OUQ framework [2] provides an approach to transform the infinite-dimensional Problem OUQ-MG into a tractable finite-dimensional problem. Owhadi et al. show that for a large class of OUQ problems, one can leverage Winkler's theorem of extremal measures [7]. In the context of Problem OUQ-MG, this states that the extremal measure $(\mu_{x_0} \times \mu_{y_0})^*$ is a discrete measure. Furthermore, the corresponding extremal marginal laws $\mu_{x_0}^*$ and $\mu_{y_0}^*$ are also discrete measures with each having at most $N + 1$ support points, where N is the number of linearly independent expectation constraints involving the relevant random variable. For admissible set $\mathcal{A}_1$, since there is one expectation constraint for each of random variables x_0 and y_0 , each of the extremal measures $\mu_{x_0}^*$ and $\mu_{y_0}^*$ has 2 support points and thus the product extremal measure $(\mu_{x_0} \times \mu_{y_0})^*$ has 4 two-dimensional support points. Thus, we can reduce Problem OUQ-MG to a finite-dimensional constrained optimization problem with 8 decision variables (the 4 support points and the corresponding weights). Within `OptimalUncertaintyQuantification.jl`, a user can leverage this transformation algorithm by specifying the `WinklerExtremalMeasures` keyword argument.

A further development was proposed by Stenger et al. [8, 9] leveraging the theory of canonical moments presented in [10]. This approach is applicable to cases where the only information constraints specified in the admissible set are the sequence of raw moments for each of the random variables. The canonical moment transformations involve normalizing the raw moment sequence in measure space. This transformation is applied in tandem with Winkler's theorem of extremal measures to construct a finite-dimensional unconstrained optimization problem. `OptimalUncertaintyQuantification.jl` contains a `CanonicalMoments.jl` sub-package that implements the necessary transformations; a user can leverage this transformation by specifying the `StengerCanonicalMoments` keyword argument (as done in Section V.D).

B. Reachability Analysis and Validated Solution of ODEs

The objective function of Problem OUQ-MG involves the state of the ODE at particular time slice which in turn depends on the distributions μ_{x_0} and μ_{y_0} of the initial conditions. Thus, the solution of the ODE pushes forward the distributions μ_{x_0} and μ_{y_0} to the distributions of the corresponding states at the specified time slice. The Julia `ReachabilityAnalysis.jl` [11] ecosystem provides a suite of tools to compute the set of states attainable by the ODE system from the set of initial conditions (in this case the bounds on the initial condition random variables). Furthermore, considering that the purpose of this work is for certification of safety-critical dynamical system models, we have one further requirement: We also need to account for numerical error that may be introduced by the ODE system integrator. For this, we leverage Rigorous Polynomial Approximations (RPAs) [12] and in particular Taylor Models (implemented in the `TaylorModels.jl` [13] package) for validated solution of ODEs. Within `OptimalUncertaintyQuantification.jl`, the user may choose any of the validated ODE solution algorithms provided by the Julia Reachability ecosystem. For example, in Section V.D we choose the TMJets algorithm which constructs a Taylor model comprising a 4^{th} -order polynomial in time and 2^{nd} -order polynomial in space.

C. Integration with Global Optimization Solvers

The transformed optimization problem can be formulated using either the `Optimization.jl` or `JuMP.jl` modeling languages. The user may choose a suitable solver for the problem available from these ecosystems. For example, in Section V.D, we use a solver from the `OptimizationBBO.jl` package.

In addition, we also developed `SpatialBranchAndBound.jl` - a solver for determining the global optimum of a nonconvex Nonlinear Program to within guaranteed bounds. This solver formulates and solves a sequence of upper bounding and lower bounding subproblems that bracket the global optimum. For upper bounding the global optimum (of a minimization problem), any local optimization solver may be used as the objective value evaluated at any feasible point is an upper bound. Lower bounding involves implementing a convexification strategy (such as McCormick-based relaxations [14, 15] or Moment Sum-Of-Squares formulations [16, 17]) or using Interval Arithmetic for range-bounding [18] all of which can be chosen conveniently in `SpatialBranchAndBound.jl`.

D. User Interface

```
1 using SymbolicOUQ
2 using Symbolics, ModelingToolkit, OrdinaryDiffEq, Optimization, OptimizationBBO
3 using ReachabilityAnalysis, TaylorModels
4 using ModelingToolkit: t_nounits as t, D_nounits as D
5
6 # 1. Define the Initial Value Problem using ModelingToolkit.jl
7 var = @variables begin
8     x(t),
9     y(t)
10 end
11
12 par = @parameters begin
13     d = -0.5
14 end
15
16 model_eqns = [
17     D(x) ~ -y - 1.5 * x^2 - 0.5 * x^3 + d
18     D(y) ~ 3 * x - y
19 ]
20
21 @named model = ODESystem(model_eqns, t, var, par)
22
23 # 2. i. Specify the random variables
24 rand_vars = @random_variables begin
25     Independent(x0, bounds = (0.8, 1.2))
26     Independent(y0, bounds = (0.8, 1.2))
27 end
28
29 #ii. Specify the information constraints (in this case 1st moments of the random variables) and
30     the admissible set
31 information_constraints = [
32     E(x0) ~ 1.0
```

```

32 E(y0) ~ 1.0
33 E(x0^2) ~ 1.0 # Only for admissible set A4
34 E(y0^2) ~ 1.0 # Only for admissible set A4
35 ]
36
37 #iii). Specify the observable of interest. In this case, it is the probability that state y at
    time t = 2.5 s is greater than -3.1 (the unsafe region)
38
39 objective = P(y[2.5] <= -3.1)
40
41 #iv). Choose the reduction algorithm, put everything together
42 ouq_sys = OUQSystem(; objective, admissible_set, reduction_alg = StengerCanonicalMoments(),
    timespan => (0.0, 5.0), u0_map = Dict{x => x0, y => y0})
43
44 # v). Choose an ODE bounding algorithm from ReachabilityAnalysis.jl and TaylorModels.jl.
45 # We use the TMJets algorithm which creates a Taylor model composed of polynomials of order 4 in
    time and order 2 in space. Choose the optimization modeling language (Optimization.jl or
    JuMP.jl)
46
47 ODE_bounding_alg = TMJets(abstol=1e-4, orderT=4, orderQ=2, adaptive=true)
48 ouq_prob = OUQProblem(ouq_sys, OptimizationModel(), Oracle(); ODE_bounding_alg)
49
50 # vi). Solve the OUQ problem using an appropriate solver. In this case, we
51 sol = solve(ouq_prob.optim_model, BBO_adaptive_de_rand_1_bin_radiuslimited())

```

Listing 1 User Interface for formulating the OUQ problem of the Moore-Greitzer model

VI. Preliminary Results and Discussion

Fig.(1) presents the reachable sets in phase space (left) and in time (right) starting from the specified initial condition set. Fig.(2) is a contour plot showing the value of the observable state y at time $t = 2.5$ s.

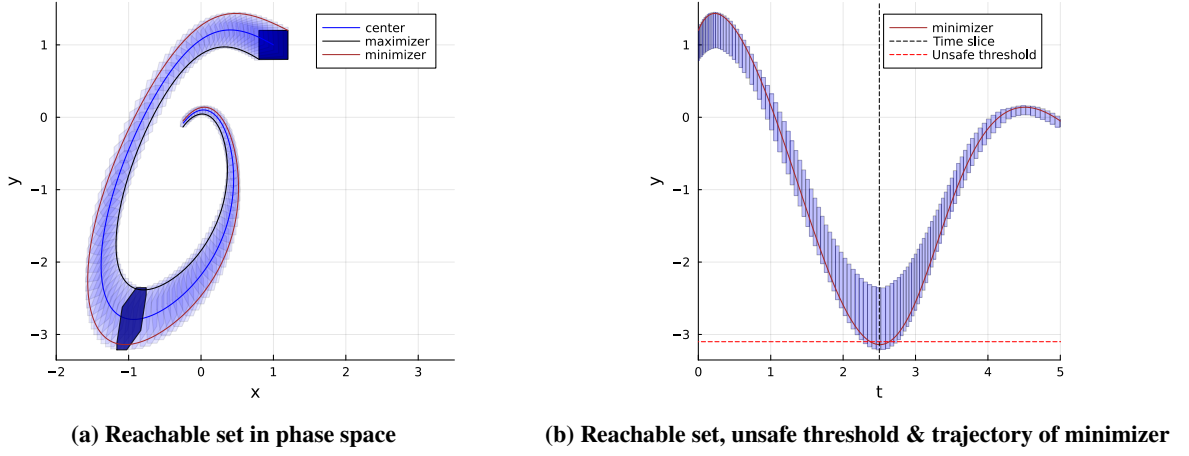


Fig. 1 Results for the OUQ problem of the Moore-Greitzer model.

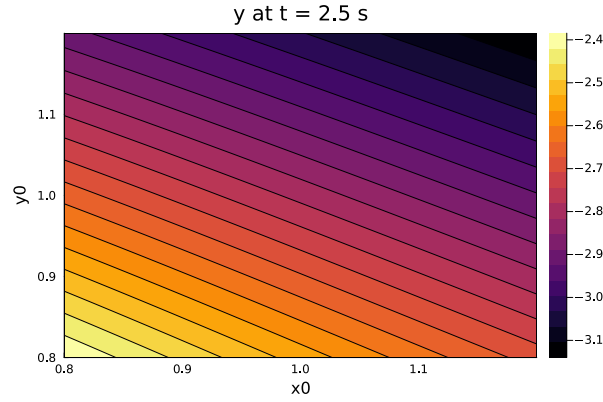


Fig. 2 Contour plot of observable state y attained at different initial conditions

Note that both the figures above are attained solely using Julia reachability analysis tools do not make use of the additional information on the moments of the initial condition random variables.

Table 1 presents the results of solving OUQ Problem OUQ-MG to compute the maximum probability of entering the unsafe region. We note that with admissible set $\mathcal{A}_1$, the maximum probability is above the safety threshold thus the model is decertified. The modeler may then proceed to iteratively change the design: Consider the case of modifying the design such that (experimental) data now produce tighter bounds on the uncertain random variables x_0 and y_0 as shown in admissible sets $\mathcal{A}_2$ and $\mathcal{A}_3$ below. We observe that the design with bounds $\mathcal{A}_2$ still yields an unacceptably high maximum probability yet design $\mathcal{A}_3$ has a maximum probability lower than the threshold and is thus certified. Finally, we consider design $\mathcal{A}_4$ where the random variables have the same bounds as $\mathcal{A}_1$ but in this case 2^{nd} -order moment information is available. We observe that this additional information sufficiently constrains the space of consistent product measures such that the worst-case probability is now lower than the threshold and the design can be certified.

Table 1 Results of solving the OUQ Problem for certification of the Moore-Greitzer model (with a maximum allowable threshold of 2%)

Admissible set	Maximum Probability of Entering Unsafe Region (%)	Certified / Decertified
$\mathcal{A}_1$	31.3	Decertified
$\mathcal{A}_2$	26.5	Decertified
$\mathcal{A}_3$	0.0	Certified
$\mathcal{A}_4$	0.0	Certified

where:

$$\mathcal{A}_2 = \left\{ \mu_{x_0} \times \mu_{y_0} \left| \begin{array}{l} (x_0, y_0) \in [0.82, 1.18]^2 \subseteq \mathbb{R}^2, \\ \mathbb{E}[x_0 \mid \mu_{x_0}] = 1.0, \\ \mathbb{E}[y_0 \mid \mu_{y_0}] = 1.0 \end{array} \right. \right\} \quad (\text{A2})$$

$$\mathcal{A}_3 = \left\{ \mu_{x_0} \times \mu_{y_0} \left| \begin{array}{l} (x_0, y_0) \in [0.84, 1.16]^2 \subseteq \mathbb{R}^2, \\ \mathbb{E}[x_0 \mid \mu_{x_0}] = 1.0, \\ \mathbb{E}[y_0 \mid \mu_{y_0}] = 1.0 \end{array} \right. \right\} \quad (\text{A3})$$

$$\mathcal{A}_4 = \left\{ \mu_{x_0} \times \mu_{y_0} \left| \begin{array}{l} (x_0, y_0) \in [0.8, 1.2]^2 \subseteq \mathbb{R}^2, \\ \mathbb{E}[x_0 \mid \mu_{x_0}] = 1.0, \\ \mathbb{E}[y_0 \mid \mu_{y_0}] = 1.0 \\ \mathbb{E}[x_0^2 \mid \mu_{x_0}] = 1.0, \\ \mathbb{E}[y_0^2 \mid \mu_{y_0}] = 1.0 \end{array} \right. \right\} \quad (\text{A4})$$

A. Features Under Development And Future Work

We list the following next steps and features under development:

- A benchmark suite with a variety of examples relevant to problems in structural engineering, aerospace and robotics.
- Further robustness improvements to `SpatialBranchAndBound.jl` to enable attainment of guaranteed global optima for a wide range of OUQ problems.
- Functionality for sensitivity analysis of the attained worst-case probabilities to model parameters.
- Development and integration with GPU-compatible optimization solvers such as [19] and [20].
- Transformations and optimization formulations that exploit model structure (such as the polynomial structure of Taylor models).
- Incorporate the use of deep learning-based uncertainty quantification schemes (such as [21] together with a data ingestion engine for automatic incorporation of experimental data.

VII. Conclusion

This work presents a Julia tool for certification of dynamical systems models by solving an end-to-end distributionally robust uncertainty quantification problem. We demonstrated the use case with a relevant benchmark of the Moore-Greitzer model which shows its practical relevance to safety-critical aerospace applications. We showed how this tool can be used by model-based system designers to iteratively meet safety and regulatory certification targets.

References

- [1] Owhadi, H., Scovel, C., and Sullivan, T., “On the brittleness of Bayesian inference,” *siam REVIEW*, Vol. 57, No. 4, 2015, pp. 566–582.
- [2] Owhadi, H., Scovel, C., Sullivan, T. J., McKerns, M., and Ortiz, M., “Optimal uncertainty quantification,” *Siam Review*, Vol. 55, No. 2, 2013, pp. 271–345.
- [3] Aylward, E. M., Parrilo, P. A., and Slotine, J.-J. E., “Stability and robustness analysis of nonlinear systems via contraction metrics and SOS programming,” *Automatica*, Vol. 44, No. 8, 2008, pp. 2163–2170.
- [4] SciML Development Team, “Overview of Julia’s SciML,” , 2024. URL <https://docs.sciml.ai/Overview/stable/>, accessed 21 May 2025.
- [5] Gowda, S., Ma, Y., Cheli, A., Gwóźdz, M., Shah, V. B., Edelman, A., and Rackauckas, C., “High-performance symbolic-numerics via multiple dispatch,” *ACM Communications in Computer Algebra*, Vol. 55, No. 3, 2022, pp. 92–96.
- [6] Ma, Y., Gowda, S., Anantharaman, R., Laughman, C., Shah, V., and Rackauckas, C., “Modelingtoolkit: A composable graph transformation system for equation-based modeling,” *arXiv preprint arXiv:2103.05244*, 2021.
- [7] Winkler, G., “Extreme points of moment sets,” *Mathematics of Operations Research*, Vol. 13, No. 4, 1988, pp. 581–587.
- [8] Stenger, J., Gamboa, F., Keller, M., and Iooss, B., “Canonical Moments for Optimal Uncertainty Quantification on a Variety,” *Geometric Science of Information: 4th International Conference, GSI 2019, Toulouse, France, August 27–29, 2019, Proceedings* 4, Springer, 2019, pp. 571–578.
- [9] Stenger, J., Gamboa, F., Keller, M., and Iooss, B., “Optimal Uncertainty Quantification of a risk measurement from a thermal-hydraulic code using Canonical Moments,” *International Journal for Uncertainty Quantification*, Vol. 10, No. 1, 2020.
- [10] Dette, H., and Studden, W. J., *The theory of canonical moments with applications in statistics, probability, and analysis*, John Wiley & Sons, 1997.
- [11] Bogomolov, S., Forests, M., Frehse, G., Potomkin, K., and Schilling, C., “JuliaReach: a toolbox for set-based reachability,” *Proceedings of the 22nd ACM International Conference on Hybrid Systems: Computation and Control*, 2019, pp. 39–44.
- [12] Joldes, M. M., “Rigorous polynomial approximations and applications,” Ph.D. thesis, Ecole normale supérieure de lyon-ENS LYON, 2011.
- [13] Benet, L., Forests, M., Sanders, D., and Schilling, C., “TaylorModels.jl: Taylor models in Julia and their application to validated solutions of ODEs,” *SWIM*, 2019, pp. 15–16.
- [14] Mitsos, A., Chachuat, B., and Barton, P. I., “McCormick-based relaxations of algorithms,” *SIAM Journal on Optimization*, Vol. 20, No. 2, 2009, pp. 573–601.
- [15] Scott, J. K., Stuber, M. D., and Barton, P. I., “Generalized mccormick relaxations,” *Journal of Global Optimization*, Vol. 51, No. 4, 2011, pp. 569–606.
- [16] Lasserre, J. B., “Global optimization with polynomials and the problem of moments,” *SIAM Journal on optimization*, Vol. 11, No. 3, 2001, pp. 796–817.
- [17] Weisser, T., Legat, B., Coey, C., Kapelevich, L., and Vielma, J. P., “Polynomial and Moment Optimization in Julia and JuMP,” *JuliaCon*, 2019. URL <https://pretalx.com/juliacon2019/talk/QZBKAU/>.
- [18] Sanders, D. P., and Benet, L., “IntervalArithmetic.jl,” , 2014. <https://doi.org/10.5281/zenodo.3336308>, URL <https://github.com/JuliaIntervals/IntervalArithmetic.jl>.
- [19] Utkarsh, Dixit, V. K., Samaroo, J., Pal, A., Edelman, A., Rackauckas, C. V., et al., “Efficient GPU-Accelerated Global Optimization for Inverse Problems,” *ICLR 2024 Workshop on AI4DifferentialEquations In Science*, ???

- [20] Kang, S., Xu, X., Sarva, J., Liang, L., and Yang, H., “Fast and certifiable trajectory optimization,” *arXiv preprint arXiv:2406.05846*, 2024.
- [21] Sun, X., and Liu, B., “A learning-based optimal uncertainty quantification method and its application to ballistic impact problems,” *Mechanics of Materials*, Vol. 184, 2023, p. 104727.